

Day 7 – Advanced Log Analysis & Event Interpretation

1. Key Log Fields

Every security log entry contains a set of core fields. Learning to read these fields quickly is fundamental to SOC triage. Before deciding whether activity is malicious, an analyst should establish the basic facts of the event.

Field	What it tells you
Timestamp	Exactly when the event occurred. It is essential for building a timeline and correlating activity across systems.
Source IP	Where the connection or request originated, such as an internal host, external/public IP, or VPN endpoint.
User/Account	Which identity performed or was targeted by the action. This helps identify targeted accounts and possible privilege escalation.
Event Type	The category of activity recorded, such as logon, logoff, process creation, or privilege use.
Process/Service	The application, service, or executable involved. This can help identify unusual or unauthorized software.

2. Windows Security Event IDs

Windows records security-relevant activity in the Security Event Log. Each event is identified by a numeric Event ID. The following IDs are commonly useful during SOC triage:

Event ID	Meaning	Why it matters
4624	Successful Logon	Confirms successful authentication. It is normal in isolation, but may be suspicious when it follows repeated failures or occurs at an unusual time or location.
4625	Failed Logon	A single failure can be normal, but a rapid burst from one source can indicate brute-force or credential-guessing activity.
4672	Special Privileges Assigned to New Logon	Indicates that special privileges were assigned to a logon session. It deserves attention when it follows a suspicious logon.
4688	Process Creation	Records a new process. It is useful for identifying unusual executables or suspicious process chains.

3. Identifying Suspicious Patterns

- Multiple failed login attempts — especially many attempts in a short time window against the same account — can indicate password guessing or brute force.
- Unusual source IP — for example, an unfamiliar network, unexpected geographic origin, or a known malicious IP range.
- Successful login after repeated failures — a strong indicator that a credential attack may have succeeded, particularly when the source and target remain the same.
- Suspicious process execution — for example, PowerShell with encoded commands, unexpected administrative tools, or processes launched immediately after an unusual login.

4. Log Investigation – Worked Example

The following is a sample scenario: repeated failed logins followed by a successful login and a process launched immediately afterward.

09/09/2026 09:14:02	EventID=4625	Account=jsmith	SourceIP=185.212.44.18	Workstation=WIN-SRV01	Status=Bad password
09/09/2026 09:14:05	EventID=4625	Account=jsmith	SourceIP=185.212.44.18	Workstation=WIN-SRV01	Status=Bad password
09/09/2026 09:14:09	EventID=4625	Account=jsmith	SourceIP=185.212.44.18	Workstation=WIN-SRV01	Status=Bad password
09/09/2026 09:14:47	EventID=4625	Account=jsmith	SourceIP=185.212.44.18	Workstation=WIN-SRV01	Status=Bad password
09/09/2026 09:15:03	EventID=4624	Account=jsmith	SourceIP=185.212.44.18	Workstation=WIN-SRV01	LoginType=3
09/09/2026 09:15:10	EventID=4672	Account=jsmith	Privileges=SeDebugPrivilege	Workstation=WIN-SRV01	
09/09/2026 09:15:22	EventID=4688	Account=jsmith	NewProcess=powershell.exe	CommandLine=-enc SQBFAFgA...	Workstation=WIN-SRV01

What happened?

Four failed login attempts (4625) against jsmith occurred within roughly 45 seconds, followed by a successful login (4624), a 4672 special-privilege event, and a 4688 PowerShell process creation event.

Who was targeted?

The account jsmith on host WIN-SRV01.

Where did the activity originate?

The activity originated from external source IP 185.212.44.18. In a real investigation, the IP should be checked against threat-intelligence and geolocation sources.

Is it suspicious?

Yes. The sequence of repeated failures followed by a success from the same source is suspicious. The subsequent privileged logon activity and encoded PowerShell command increase concern.

What attack could it indicate?

It could indicate a brute-force or credential-stuffing attack against a remotely accessible service, followed by post-exploitation activity using an obfuscated PowerShell command.

What is the risk?

High. If the credentials were compromised and privileged code execution occurred on a server, the attacker could potentially perform lateral movement, steal data, or deploy ransomware.

5. SOC Investigation Process

- 1 **Log** — The SIEM ingests the 4625, 4624, 4672, and 4688 events from WIN-SRV01.
- 2 **Identify Pattern** — A correlation rule can alert when multiple failed logons are followed by a successful logon from the same source within a short window.
- 3 **Investigate** — The analyst reviews the complete event timeline, checks source-IP reputation, and examines the PowerShell command line.

- 4 **Determine Threat** — The activity is treated as a likely successful credential attack followed by suspicious post-logon activity.
- 5 **Assess Risk** — The incident is rated High because privileged access and code execution appear on a server.
- 6 **Respond** — Reset or disable the compromised account, isolate the host where appropriate, block the malicious source if confirmed, and escalate for deeper forensic investigation.

6. SOC Role Mapping

Tier	Responsibility	Role in this example
L1	Monitor and identify suspicious activity	Notifies the correlation alert for repeated failed logins followed by a success and opens an initial ticket.
L2	Investigate and confirm incidents	Reviews the log timeline, validates the suspicious sequence and PowerShell activity, and escalates as needed.
L3	Threat hunting and improve detection	Performs deeper analysis, investigates the PowerShell payload, hunts for similar activity, and tunes detection rules.

Assignment – Sample Security Analysis Report

1. Attack Identification

Likely brute-force login activity against the account jsmith on WIN-SRV01, followed by suspicious post-logon activity involving an obfuscated PowerShell command.

2. Evidence from Logs

- Four consecutive 4625 failed-logon events for jsmith from 185.212.44.18 within roughly 45 seconds.
- A 4624 successful network logon (Logon Type 3) immediately following the failures from the same IP.
- A 4672 event showing special privileges assigned to the same account shortly after the successful logon.
- A 4688 event showing powershell.exe launched with an -enc (Base64-encoded) command.

3. Investigation Findings

The timeline shows a suspicious repeated-failure-to-success pattern consistent with a possible credential attack. The source IP is external and should be validated against the organisation's expected address space and threat-intelligence data. The privilege assignment and encoded PowerShell command immediately after the login strongly increase the likelihood of post-compromise activity. The exact intent of the command cannot be confirmed from the truncated sample alone.

4. Risk Level

High — the combination of a possible credential compromise, privileged access, and code execution on a server presents significant risk of lateral movement, data exfiltration, persistence, or ransomware deployment.

5. Recommended Response

- Immediately disable or reset the jsmith account and force a password change.
- Isolate WIN-SRV01 from the network when operationally appropriate and begin forensic review.
- Block or investigate 185.212.44.18 after validating that it is malicious and not a legitimate source.
- Decode and analyse the PowerShell command line in a safe environment to identify its intent and indicators of compromise (IOCs).
- Search other logs and hosts for the same source IP, account, and related indicators.
- Escalate to L2/L3 for full incident response and formal incident declaration if malicious activity is confirmed.

6. Prevention Techniques

- Enforce appropriate account lockout or throttling controls to reduce brute-force attempts.
- Require multi-factor authentication (MFA) for remote and network logons where supported.
- Restrict and monitor internet exposure of remote-access services such as RDP and VPN.
- Enable PowerShell logging and appropriate application controls, and alert on suspicious encoded-command usage.

- Use threat-intelligence feeds to identify known malicious source IPs.
- Regularly review and tune correlation rules so brute-force-to-success patterns generate high-priority alerts.